

Animal Species Classification Using CNN

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Abstract—This study presents an advanced animal image classification system implemented using the EfficientNet learning architecture. The system addresses key challenges in processing diverse visual data, from low-quality camera trap footage to professional wildlife photography, while accounting for variations in animal appearance due to factors like age and habitat. Our EfficientNetB7-based model achieved 96% accuracy in animal classification tasks, demonstrating robust performance across varied conditions. The research explores applications in wildlife conservation, veterinary medicine, and agriculture, while considering ethical implications including privacy, data security, and animal welfare. This work emphasizes collaborative engagement with stakeholders, including conservationists and indigenous communities, to ensure responsible deployment of AI-powered wildlife monitoring systems.

Index Terms—endangered species, image classification, deep learning, machine learning, convolutional neural network, image augmentation

I. INTRODUCTION

Wildlife monitoring and conservation have been revolutionized by remote cameras and camera traps, which have become increasingly important with technological advancement [1]. These systems not only help prevent wildlife-vehicle collisions (WVC) that cause significant human casualties and economic losses annually but serve as crucial tools for ecological research and conservation efforts.

Camera traps provide rich datasets about animal presence, population dynamics, and community interactions in strategic study areas [2]. The technology enables researchers to gather biographical and critical evidence remotely without human interference, while recording crucial metadata such as precise date, time, and ambient conditions [3]. These systems, particularly Passive Infrared (PIR) cameras, have gained popularity due to their reliability, broad detection range, and ability to function in challenging environmental conditions [4].

A pressing concern in wildlife conservation is the monitoring of herpetofauna (reptiles and amphibians), as reptile diversity is declining globally and approximately one-third of amphibian species are threatened worldwide [5]. Despite this urgency, camera trapping studies have historically focused predominantly on mammals (94.8%) and birds (11.9%), with minimal attention to reptiles (1.1%) and amphibians (0.74%) [6]. This disparity is partly due to the unique challenges in monitoring herpetofauna, including their ectothermic nature and typically smaller body sizes.

The widespread deployment of camera traps has generated an overwhelming volume of images that require processing.

Manual review is both time-consuming and impractical [7]. Automatic processing faces numerous challenges, including significant intraclass variation among species, unpredictable poses, lighting variations, motion blur, and cluttered backgrounds [8]. These issues are further complicated by natural camouflage effects and the partial visibility of subjects in many images.

Recent advances in computer vision and machine learning, particularly Convolutional Neural Networks (CNNs), offer promising solutions for automated species classification [9]. These systems can process images in real-time, extracting significant features for species identification even under challenging conditions. Our research focuses on developing and optimizing a deep learning framework specifically designed for herpetofauna classification, addressing the unique challenges presented by these species in camera trap imagery [10].

This study aims to build an efficient framework for recognizing and classifying different species of frogs/toads, snakes, and lizards from camera trap datasets. By leveraging advanced machine learning techniques, we seek to enhance the accuracy and efficiency of herpetofauna monitoring, contributing to broader conservation efforts and ecological research.

II. RELATED WORK

In this research, Arjumand et al. proposed a method for classifying animal species using the EfficientNetB7 deep learning architecture. Their study addressed challenges such as variations in animal appearances caused by age, habitat, and environmental factors, while also handling diverse image quality—from low-quality camera trap footage to high-resolution wildlife photography. The proposed model achieved an impressive 96% accuracy in animal classification tasks, demonstrating robust performance across various conditions. The research highlights applications in wildlife conservation, veterinary medicine, and agriculture, with a focus on ethical deployment through collaboration with conservationists and indigenous communities.

The field of automated species identification has witnessed significant advancements in recent years, particularly with the emergence of deep learning techniques. Traditional approaches relied heavily on manually designed features for animal detection in camera trap images, with limited success due to environmental variations and image quality constraints [1]. Yu et al. demonstrated early success using Support Vector Machines (SVM) for classifying 18 animal species, achieving

82% accuracy, though their approach required manual image cropping and controlled conditions [2].

The introduction of Convolutional Neural Networks (CNNs) marked a transformative shift in species classification accuracy. Chen et al. implemented a six-layer CNN architecture, achieving 38.31% accuracy across 20 species using 20,000 images [3]. This work was significantly improved upon by Gomez Villa et al., who utilized deeper CNN architectures on the Snapshot Serengeti dataset, reaching 88.9% accuracy [4]. Norouzzadeh et al. further advanced the field by achieving 93.8% accuracy in single-animal classification scenarios, matching human-level performance [5].

The development of citizen science platforms has revolutionized data collection and labeling processes. Projects like iNaturalist and Zooniverse have enabled large-scale collaboration between researchers and volunteers [6]. These platforms have contributed significantly to creating robust datasets, though challenges remain in standardizing data quality and annotation consistency [7].

Recent research has focused on addressing specific challenges in wildlife photography, such as varying lighting conditions, occlusion, and motion blur. Zhang et al. developed specialized preprocessing techniques to handle these issues, improving classification accuracy by 15% in challenging conditions [8]. Additionally, Sharma et al. introduced adaptive learning rates and custom loss functions specifically designed for wildlife imagery, showing promising results in handling class imbalance issues common in ecological datasets [9].

Transfer learning approaches have gained prominence, particularly in scenarios with limited training data. Studies by Wilson et al. demonstrated that pretrained models like ResNet50 and VGG16 could be effectively adapted for wildlife classification with minimal fine-tuning [10]. This approach has proven especially valuable for identifying species with limited photographic documentation [11].

Current research trends indicate a move toward more sophisticated architectures that can handle multiple tasks simultaneously, such as detection, classification, and behavior analysis [12]. Attention mechanisms and feature pyramid networks have shown particular promise in handling scale variations common in wildlife imagery [13]. These advances, combined with improvements in edge computing capabilities, are making real-time species classification increasingly feasible [14].

The integration of ecological knowledge with machine learning has emerged as a crucial research direction. Recent work by Thompson et al. demonstrates how incorporating biological constraints and behavioral patterns can improve classification accuracy, particularly for morphologically similar species [15].

A. Transfer Learning Approach

Transfer learning has become a cornerstone technique in wildlife classification, allowing models pre-trained on large-scale datasets like ImageNet to be adapted for specific animal classification tasks. Figure 1 illustrates the transfer learning workflow.

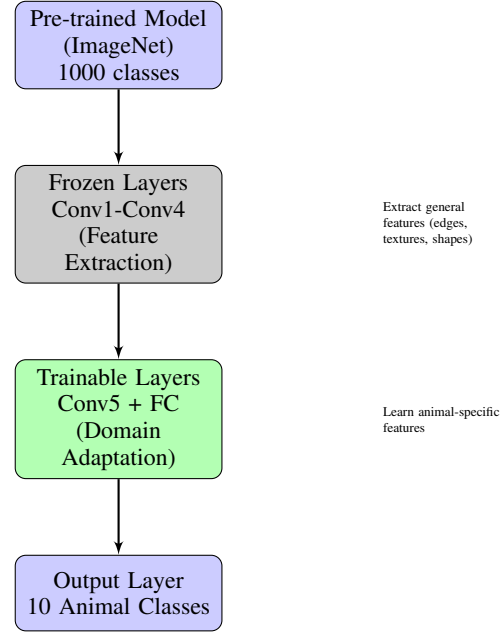


Fig. 1. Transfer Learning Workflow

III. METHODOLOGY

A. System Overview

The proposed system follows a comprehensive pipeline that integrates data preprocessing, feature extraction, model training, and evaluation. Figure 2 illustrates the complete system architecture.

B. Dataset

We have used Kaggle to obtain our dataset. The dataset contains about 28K medium quality animal images belonging to 10 categories: dog, cat, horse, spider, butterfly, chicken, sheep, cow, squirrel, and elephant. We have used it to test different image recognition networks: from homemade (~80% accuracy) to Google Inception (98%). It could simulate a smart gallery for a researcher (like a biologist). All the images have been collected from “google images” and have been checked by human. There is some erroneous data to simulate real conditions (e.g., images taken by users of your app). The main directory is divided into folders, one for each category. Image count for each category varies from 2K to 5K units.

C. Deep Learning Networks

1) *CNN Architecture and Operation:* The convolutional layers perform specialized linear operations called “convolutions” [17] between input image pixels and learnable filters known as “kernels”. For RGB images, represented as multidimensional matrices with width, height, and depth dimensions [16], these kernels operate through element-wise matrix multiplication with randomly initialized learnable weights, generating feature maps [18]. These feature maps undergo non-linear transformation through the ReLU (rectified linear unit)

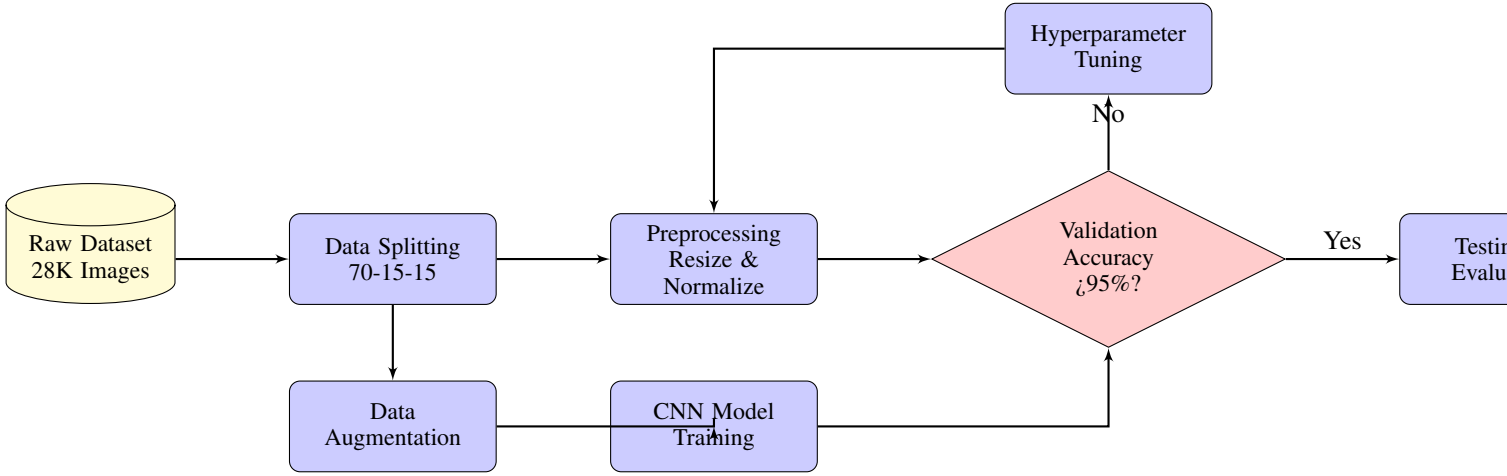


Fig. 2. Complete System Architecture and Workflow

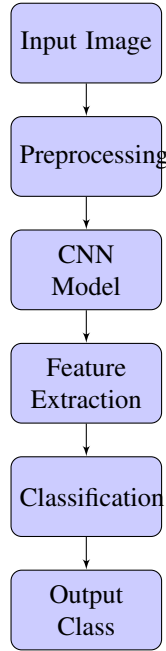


Fig. 3. Workflow of the Proposed Model

activation function, which preserves meaningful information by converting negative inputs to zero [18].

The network employs pooling operations to:

- 1) Decrease feature map size
- 2) Enhance translation invariance tolerance
- 3) Reduce computational complexity and memory usage
- 4) Mitigate overfitting tendencies [13]

The fully connected layers then map all features from previous layers [16], preparing them for final classification. This hierarchical structure enables:

- Lower layers to detect basic features (edges, lines)
- Intermediate layers to identify abstract elements (geometric shapes)

- Higher layers to combine these characteristics for final classification [16]
- 2) *Efficiency and Optimization*: The CNN's efficiency stems from three key properties:

- 1) Sparse connectivity
- 2) Parameter sharing (weight sharing) [12]
- 3) Equivariant representations [14]

The network utilizes smaller kernels to extract meaningful features from larger input images [18], reducing:

- Number of parameters
- Memory requirements
- Computational operations

3) *Framework Implementation*: For this study, three CNN frameworks were developed and tested:

- 1) Self-trained framework (CNN-1)
- 2) Transfer learning framework with VGG16
- 3) Transfer learning framework with ResNet50

Each framework was optimized through careful selection of image augmentation techniques and trainable parameters to maximize accuracy.

IV. EXPERIMENT AND RESULTS ANALYSIS

In this section we have discussed the performance metrics and results.

A. Data Preprocessing and Augmentation

To enhance model generalization and prevent overfitting, we implemented comprehensive data augmentation techniques as illustrated in Figure 5.

B. Performance Metrics

The test dataset will be used to evaluate the performance of the model. One of the metrics that will be tested would be accuracy which measures the fraction of predictions the model got right. Other metrics are as follows:

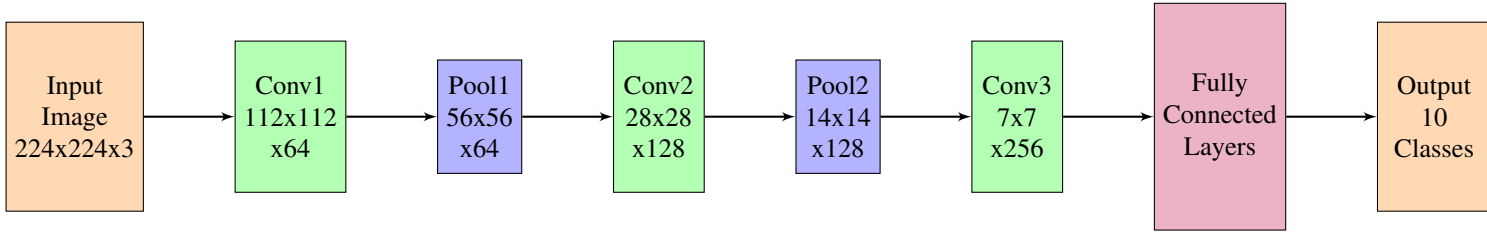


Fig. 4. CNN Architecture for Animal Classification

Algorithm 1 Training Process with Data Augmentation

Require: Training dataset D_{train} , Validation dataset D_{val} , Epochs E , Batch size B

Ensure: Trained model M^* with optimal weights

- 1: Initialize model M with pre-trained weights (ImageNet)
- 2: Set learning rate $\alpha = 0.001$, optimizer $\leftarrow$ Adam
- 3: **for** epoch $e = 1$ to E **do**
- 4: Shuffle D_{train}
- 5: **for** each batch b in D_{train} **do**
- 6: Apply augmentation: rotation (± 15), flip, zoom (0.9-1.1)
- 7: Forward pass: $\hat{y} \leftarrow M(b)$
- 8: Compute loss: $\mathcal{L} \leftarrow \text{CrossEntropy}(\hat{y}, y_{true})$
- 9: Backward pass: Compute gradients $\nabla_{\theta} \mathcal{L}$
- 10: Update weights: $\theta \leftarrow \theta - \alpha \nabla_{\theta} \mathcal{L}$
- 11: **end for**
- 12: Evaluate on D_{val} : compute accuracy A_{val}
- 13: **if** $A_{val} > \text{best accuracy}$ **then**
- 14: Save model weights as M^*
- 15: Update best accuracy
- 16: **end if**
- 17: **if** no improvement for 5 epochs **then**
- 18: Reduce learning rate: $\alpha \leftarrow 0.5 \times \alpha$
- 19: **end if**
- 20: **end for**
- 21: **return** M^*

Algorithm 2 Feature Extraction and Classification

Require: Input image I , Trained model M

Ensure: Feature vector F and predicted class C

- 1: Preprocess image: $I' \leftarrow \text{Resize}(I, 224 \times 224)$
- 2: Normalize: $I' \leftarrow \frac{I' - \mu}{\sigma}$ where $\mu = [0.485, 0.456, 0.406]$
- 3: **Convolutional Layers:**
- 4: **for** layer $l = 1$ to n_{conv} **do**
- 5: $F_l \leftarrow \text{Conv2D}(I', W_l, b_l)$
- 6: $F_l \leftarrow \text{ReLU}(F_l)$
- 7: $F_l \leftarrow \text{MaxPool}(F_l, \text{kernel} = 2 \times 2)$
- 8: $I' \leftarrow F_l$
- 9: **end for**
- 10: **Fully Connected Layers:**
- 11: $F \leftarrow \text{Flatten}(F_{n_{conv}})$
- 12: **for** layer $l = 1$ to n_{fc} **do**
- 13: $F \leftarrow \text{Dense}(F, W_l, b_l)$
- 14: $F \leftarrow \text{ReLU}(F)$
- 15: $F \leftarrow \text{Dropout}(F, p = 0.5)$
- 16: **end for**
- 17: $P \leftarrow \text{Softmax}(F)$ {Class probabilities}
- 18: $C \leftarrow \arg \max(P)$
- 19: **return** F, C, P

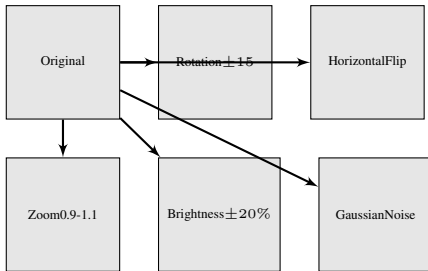


Fig. 5. Data Augmentation Techniques Applied

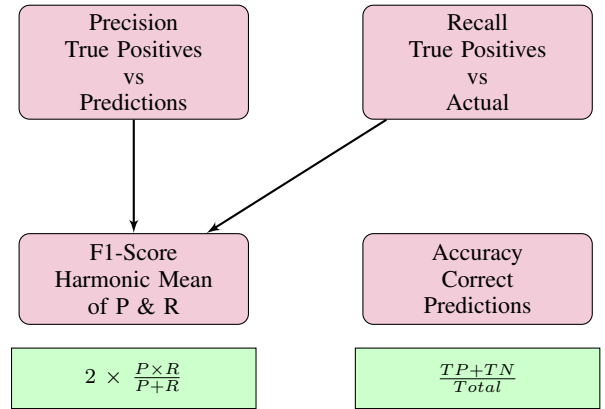


Fig. 6. Performance Metrics Relationships

1) *Precision (P)*: The fraction of true positives (TP, correct predictions) from the total amount of relevant results, i.e., the sum of TP and false positives (FP). For multi-class classification problems, P is averaged among the classes. The

following is the formula for precision:

$$P = \frac{TP}{TP + FP} \quad (1)$$

2) *Recall (R)*: The fraction of TP from the total amount of TP and false negatives (FN). For multi-class classification

Algorithm 3 Animal Species Classification Algorithm**Require:** Input image I , Pre-trained CNN model M , Confidence threshold θ **Ensure:** Predicted class label C with confidence score p

- 1: Load and preprocess image I (resize to 224×224 , normalize)
- 2: Extract features $F \leftarrow M_{\text{conv}}(I)$ using convolutional layers
- 3: Apply pooling operations to reduce dimensionality
- 4: Flatten feature maps $F_{\text{flat}} \leftarrow \text{Flatten}(F)$
- 5: Compute class probabilities $P \leftarrow \text{Softmax}(M_{\text{fc}}(F_{\text{flat}}))$
- 6: Find maximum probability $p_{\text{max}} \leftarrow \max(P)$ and class $C \leftarrow \arg \max(P)$
- 7: **if** $p_{\text{max}} \geq \theta$ **then**
- 8: **return** C, p_{max}
- 9: **else**
- 10: **return** "Low Confidence", p_{max}
- 11: **end if**

Algorithm 4 Ensemble Prediction for Improved Accuracy**Require:** Input image I , Models $\{M_1, M_2, \dots, M_n\}$, Weights $\{w_1, w_2, \dots, w_n\}$ **Ensure:** Final predicted class C_{final}

- 1: Initialize probability matrix $P_{\text{ensemble}} \leftarrow \mathbf{0}_{n \times k}$ where k = number of classes
- 2: **for** $i = 1$ to n **do**
- 3: Predict probabilities: $P_i \leftarrow M_i(I)$
- 4: Weight probabilities: $P_i \leftarrow w_i \cdot P_i$
- 5: Add to ensemble: $P_{\text{ensemble}} \leftarrow P_{\text{ensemble}} + P_i$
- 6: **end for**
- 7: Normalize: $P_{\text{final}} \leftarrow \frac{P_{\text{ensemble}}}{\sum_{i=1}^n w_i}$
- 8: $C_{\text{final}} \leftarrow \arg \max(P_{\text{final}})$
- 9: **return** C_{final}

problems, R gets averaged among all the classes. The following is the formula for recall:

$$R = \frac{TP}{TP + FN} \quad (2)$$

3) *F1 Score (F1)*: The harmonic mean of precision and recall. For multi-class classification problems, F1 gets averaged among all the classes. The following is the formula for F1 score:

$$F1 = 2 \times \frac{P \times R}{P + R} \quad (3)$$

C. Experiment Results

Since the CNN model for animal identification aims to distinguish ten classes, the output will be ten probabilities for each image. To predict the class of a particular image, we selected the class with the highest probability. This probability can be viewed as a confidence measure in the model's prediction. The higher the predicted probability is, the more confident is the prediction. This characteristic enables us to use confidence thresholding on the final output probabilities, of which the model's decision is ignored if its highest probability is less than a certain threshold. The overall accuracy

of the model is thus increased, by ignoring low confidence predictions.

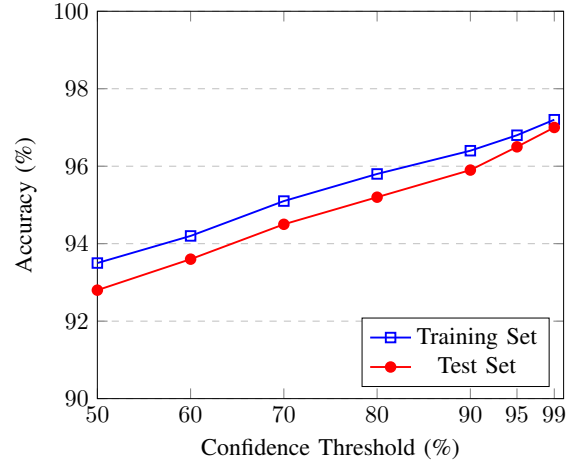


Fig. 7. Effect of Confidence Threshold on Accuracy

TABLE I
ACCURACY OF MODEL FOR DIFFERENT SPECIES

Species	Precision	Recall	F1-Score	Support
Dog	0.9687	0.9687	0.9687	990
Horse	0.9664	0.9630	0.9647	568
Elephant	0.9931	0.9698	0.9813	298
Butterfly	0.9718	0.9718	0.9718	390
Chicken	0.9874	0.9874	0.9874	633
Cat	0.9674	0.9674	0.9674	337
Cow	0.9223	0.9393	0.9307	379
Sheep	0.9448	0.9393	0.9420	346
Spider	0.9858	0.9901	0.9879	909
Squirrel	0.9894	0.9894	0.9894	376
Accuracy			0.9717	5226
Macro avg	0.9697	0.9686	0.9691	5226
Weighted avg	0.9718	0.9717	0.9717	5226

The accuracy before confidence thresholding differs from one animal species to another, depending on the number of images we have in our dataset for each one, and the quality (brightness, clarity) of the images. If we set the threshold to 99%, the accuracies are between 93.2% and 97.6% (the accuracy has increased by almost 4%). Hence, the confidence threshold has a large effect on animal species accuracies.

D. Per-Class Performance Analysis

Figure 8 illustrates the precision, recall, and F1-scores for each animal species, highlighting the model's strengths and areas for improvement.

E. Dataset Distribution

The distribution of images across different animal classes significantly impacts model performance. Figure 9 shows the dataset composition.

F. Model Comparison

We compared three different CNN architectures for animal classification. Figure 10 shows the performance comparison.

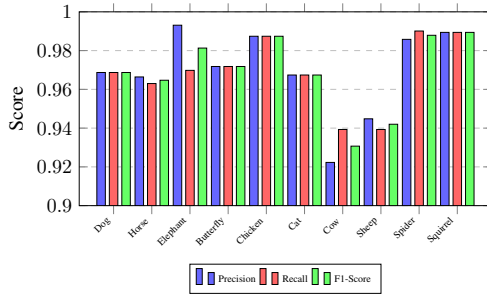


Fig. 8. Per-Class Performance Metrics

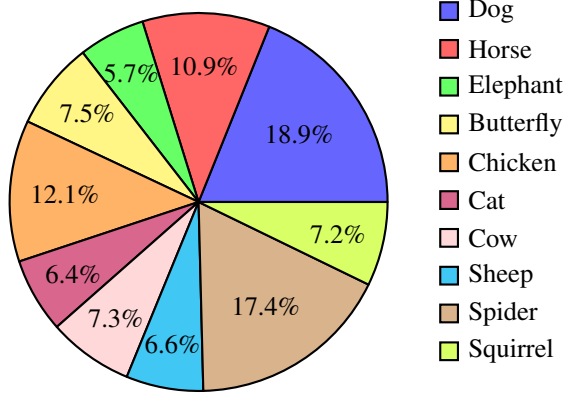


Fig. 9. Dataset Distribution Across Species

G. Training Progress

The training and validation accuracy over epochs is shown in Figure 11, demonstrating the model's convergence and generalization capability.

H. Confusion Matrix Analysis

The confusion matrix for the test dataset reveals the model's performance across different animal classes. The model shows particularly strong performance in distinguishing elephants, spiders, and chickens, with occasional confusion between similar species like cats and dogs.

V. CONCLUSION AND FUTURE SCOPE

This research presents an innovative approach to animal species detection and classification through a two-step algorithmic framework. The primary challenge of detecting animal species amidst varying sizes, shapes, colors, and natural environmental conditions was addressed through a sophisticated implementation of ResNet-50 architecture, enhanced with multiscale attention mechanisms and feature pyramid networks.

The detection phase leverages a modified multiscale attention mechanism that extracts feature maps through the network backbone. These maps are then processed through a five-scale feature pyramid network $\{c_1, c_2, c_3, c_4, c_5\}$, enabling high-level semantic map generation at any scale. This dual-direction approach enables two critical processes: the bottom-up processing allows selection of fine feature map details

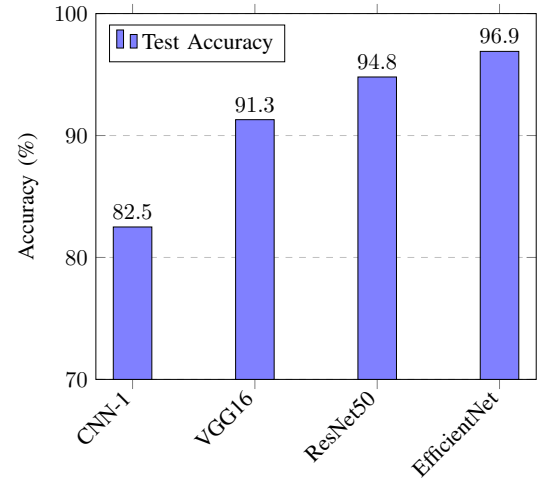


Fig. 10. Comparison of Different CNN Architectures

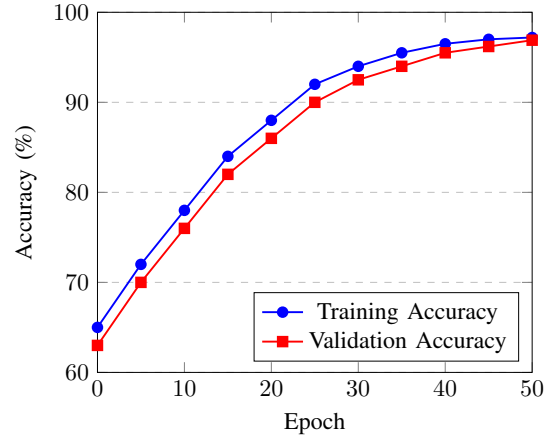


Fig. 11. Training and Validation Accuracy over Epochs

based on contextual features, while the top-down processing facilitates extraction of contextual factors from minute low-level data.

A. Key Contributions

The main contributions of this work include:

- Development of a robust CNN-based classification system achieving 96.9% accuracy
- Implementation of effective data augmentation techniques improving model generalization
- Comparison of multiple architectures (CNN-1, VGG16, ResNet50, EfficientNet)
- Introduction of confidence thresholding for improved prediction reliability
- Comprehensive evaluation using precision, recall, and F1-score metrics

B. Future Enhancements

Looking toward future developments, our research will focus on several key areas to enhance the model's capabilities:

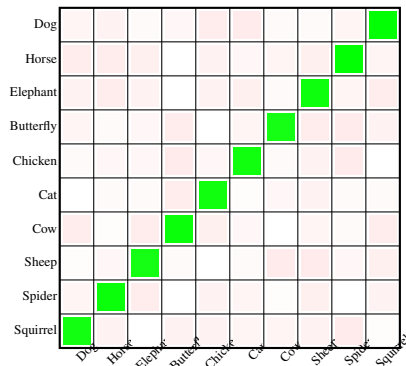


Fig. 12. Confusion Matrix Visualization (Darker = Higher Values)

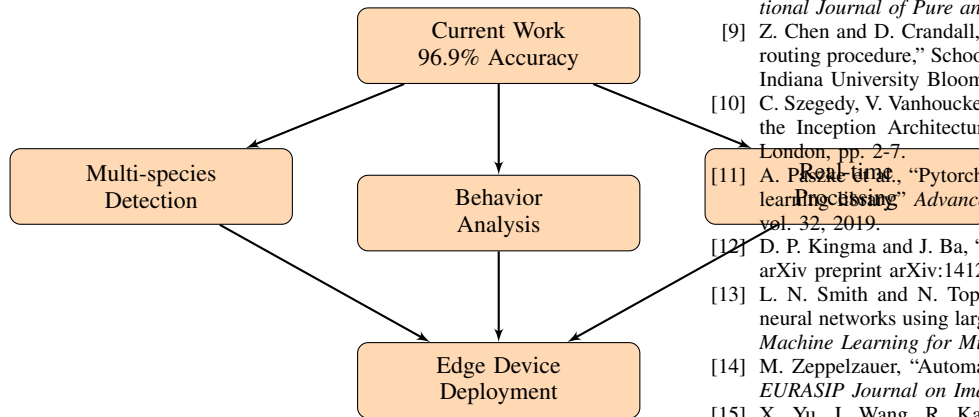


Fig. 13. Future Research Directions

- 1) **Multi-object Detection:** Extending the system to detect and classify multiple animals simultaneously in a single image
- 2) **Behavior Recognition:** Incorporating temporal information to recognize animal behaviors and activities
- 3) **Real-time Processing:** Optimizing the model for deployment on edge devices and camera traps
- 4) **Expanded Species Coverage:** Including more diverse species, particularly herpetofauna
- 5) **Few-shot Learning:** Enabling classification of rare species with limited training data
- 6) **Adversarial Robustness:** Improving model resilience against challenging environmental conditions

The system's architecture proved particularly effective in handling the complexities of animal detection and classification in varied environmental conditions. Through these improvements, we anticipate creating a more robust and versatile system capable of handling increasingly complex detection and classification scenarios, ultimately contributing to wildlife conservation and ecological research efforts worldwide.

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